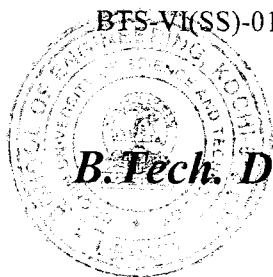


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B.Tech. Degree VI Semester Special Supplementary Examination
January 2019

CS 15-1603 CRYPTOGRAPHY AND NETWORK SECURITY
(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer ALL questions)

(10 × 2 = 20)

- I. (a) Distinguish Cryptography with Cryptanalysis.
- (b) What is an Affine Cipher? Use an Affine cipher to encrypt the message "hello" with key pair (7, 2).
- (c) Briefly explain the AES key expansion.
- (d) Describe Avalanche effect.
- (e) What are the requirements for public key cryptography?
- (f) What is Man in the middle attack in Diffie Hellman key exchange?
- (g) Explain Miller Rabin algorithm.
- (h) What are the most significant types of viruses?
- (i) What is the difference between a packet filtering router and a stateful inspection firewall?
- (j) List and briefly define the parameters that define a SSL connection state.

PART B

(4 × 10 = 40)

- II. (a) List and briefly explain the categories of passive and active security attacks. (5)
- (b) Draw and explain the model of conventional cryptosystem. (5)
- OR**
- III. (a) What is the difference between a mono alphabetic cipher and a polyalphabetic cipher? Justify your answer with suitable example. (5)
- (b) Using playfair algorithm encrypt the message "hide the gold in the tree stump" use "playfair example" as the key. (5)
- IV. Explain AES algorithm in detail. (10)
- OR**
- V. Explain various block cipher modes of operation in detail. (10)
- VI. In a public key system using RSA the attacker intercept the cipher text $c = 10$ sent to a user whose public keys $e = 5, n = 35$. What is the plain text M? (10)
- OR**
- VII. Explain Elgamal Cryptosystem in detail. (10)
- VIII. (a) Draw and explain the structure of X.509 certificate in detail. (7)
- (b) What requirements should a digital signature scheme satisfy? (3)
- OR**
- IX. (a) What is the difference between transport and tunnel mode? (5)
- (b) Briefly explain statistical anomaly detection. (5)